AQUACAST AI: Forecasting Monthly Sales for a New Beverage Product

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Business objective

A beverage company has launched a new flavoured water and needs a monthly demand forecast for the next three months to plan production and distribution. The lab tasks: preprocess with lag and seasonal features, compare Naive, ARIMA and ML models on a chronological split, forecast 3 months, score with MAPE and WAPE, plot actual vs predicted, interpret, and add an LLM interpretation.

Dataset

USDA ERS Weekly Retail Food Sales (national, by subcategory; Circana scanner data), Beverages / Water **unit sales**. Weekly data from the week ending 2018-10-07 to 2023-05-07, downloaded 2026-11-12. History before Oct 2019 uses USDA's published 'last year' and '3 years ago' fields (cross-checked within 0.03%). *Category-level proxy for the new product; historical, not live sales.*

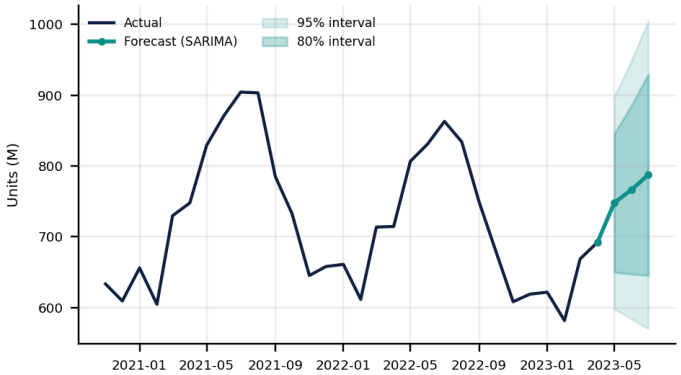
Preprocessing and method

No duplicates, gaps or missing values. A week ending D covers D-6 to D, so weeks were split across months by day; partial months were dropped (2023-05), leaving **55 months (Oct 2018 to Apr 2023)**. Features: lags 1, 2, 3 and 12, a lagged 3-month mean, month sin/cos and trend, all from past values only. Validation is chronological only: a 3-month holdout (Feb to Apr 2023) plus rolling-origin validation over 4 consecutive 3-month windows, refitting before each. MAPE = mean |A-F|/A; WAPE = 100 × Σ|A-F| / Σ|A|.

Models compared

Model	Holdout MAPE	Holdout WAPE	Rolling MAPE	Rolling WAPE
SARIMA	**2.65%**	**2.75%**	**2.29%**	**2.34%**
Random Forest	8.04%	8.00%	5.50%	5.21%
Seasonal Naive	5.03%	5.00%	5.58%	5.54%
Naive	8.02%	8.10%	11.93%	12.04%

Best model: SARIMA (1, 1, 0)×(1, 0, 0, 12), chosen for the lowest rolling-origin WAPE (2.34% against 12.04% for Naive). It models the 12-month cycle and the recent level together, while Naive repeats the last month and misses every seasonal turn.



Three-month forecast (US water category, units)

Month	Forecast	80% interval	95% interval
May 2023	747.8M	650.2M to 845.5M	598.5M to 897.2M
Jun 2023	766.5M	647.3M to 885.7M	584.2M to 948.8M
Jul 2023	787.5M	645.5M to 929.5M	570.3M to 1,004.7M
Total	**2,301.9M**		

Insights and recommendations

Seasonality is strong (strength 0.91): demand peaks in Jul (index 123) and is lowest in Feb (84). Trend is weak (strength 0.29); the last 12 months are -5.2% versus the prior 12.

• Plan production for 2023-05 to 2023-07 at about 2,302 million category units in total (-8.0% versus the same months last year), and scale that to the new product's expected share.

• Hold an extra buffer of about 359 million units across the three months (the gap to the 80% upper bound) to protect service in the peak season.

• Front-load distribution before Jul, the peak month, and plan promotions or capacity for the Feb trough.

LLM integration

Google Gemini (gemini-3.6-flash) receives the computed results as JSON, with an instruction to use no other numbers. It runs in the notebook (key from Colab Secrets) and in the AQUACAST AI Copilot, which interprets whatever dataset is currently loaded. Without a key, both show a labelled rule-based fallback, never fabricated AI text.

Limitations and conclusion

This is a national category proxy, not the new product's regional sales, and the data ends in May 2023. There are no price, promotion or weather drivers, and 55 months is a small sample. Even so, seasonal SARIMA forecasts unseen months to within about 2 to 3% error, a sound basis for production volume, safety buffer and distribution timing once the product's own sales replace the proxy.

Code, notebook and live app: github.com/mercydeez/atharva-soundankar-aquacast-ai-as25dxb020 · All figures come from the executed notebook (results.json).